

Global Population & Demographic Change Explorer

Interactive D3.js Visualization of Long-Term Demographic Transitions (1950 – 2023)

OFFICIAL LIVE PRODUCTION DEPLOYMENT

<https://big-data-visualization-one.vercel.app/>

Hosted on **Vercel Edge Network**
Framework: **Pure HTML5 / D3.js v7**

SOURCE DATASET

UN WPP 2024 Revision / OWID

REPOSITORY URL

github.com/TheCreativeCodeFlow/big-data-visualization

TEMPORAL COVERAGE

1950 – 2023 (74 Historical Years)

1. Executive Summary

Understanding long-term demographic trajectories is vital for global economic forecasting, public pension sustainability, healthcare infrastructure allocation, and labor supply planning. This academic project presents the design, methodology, implementation, and empirical observations of the **Global Population & Demographic Change Explorer**, an interactive data visualization platform developed strictly with native browser technologies (HTML5, CSS3, ES6 JavaScript) and **D3.js v7**.

The platform analyses official demographic statistics spanning **237 sovereign nations and territories** across **6 continents** over **74 continuous historical years (1950 to 2023)**. Core visualization fundamentals are demonstrated without high-level charting libraries: custom scale transformations (`d3.scaleBand`, `d3.scaleLinear`), stacked layout coordinate generation (`d3.stack`, `d3.area`), DOM data join lifecycles (`.join`), animated coordinate transitions (`d3.transition`), and real-time in-memory multi-key rollups across 87,690 records.

The complete project is deployed and publicly accessible to academic mentors and evaluators at: <https://big-data-visualization-one.vercel.app/>.

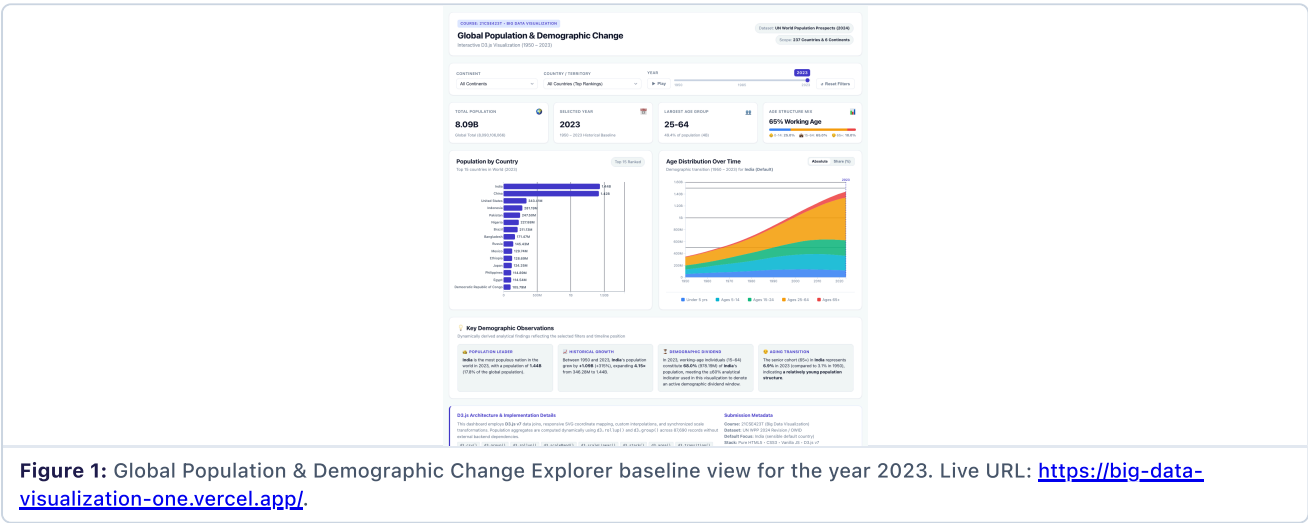


Figure 1: Global Population & Demographic Change Explorer baseline view for the year 2023. Live URL: <https://big-data-visualization-one.vercel.app/>.

2. Dataset Engineering & Preprocessing Methodology

The visualization operates on real-world demographic data compiled from the **United Nations Department of Economic and Social Affairs, Population Division (2024 Revision)**, published through *Our World in Data (OWID)*. No synthetic, simulated, or randomized values are used.

2.1 Raw Dataset Extraction & Cleaning (`scripts/preprocess.py`)

The raw source dataset (`data/raw_population_by_age.csv`) comprised 19,388 entity-year records across 8 columns. However, raw records intermingled sovereign nations with multi-national aggregate groupings (e.g., `OWID_WRL` , `UN_AFR` , Low-income countries) and lacked standardized continental classifications for individual countries.

- 1. **Entity Validation:** Filtered records to isolate 237 sovereign nations and recognized territories with standard ISO-3166-1 alpha-3 codes (plus Kosovo under `KOS`). Regional duplicates and non-geographic groupings were systematically excluded.
- 2. **Continental Mapping:** Joined all entities against the United Nations M49 regional classification reference (`data/iso_countries.csv`). Countries were classified into six continents: Africa (58 nations), Asia (51), Europe (50), North America (41), Oceania (23), and South America (14).
- 3. **Schema Normalization:** Pivoted wide age columns into a clean long-format structure (`data/demographic.csv`) supporting five discrete cohorts: *Under 5*, *5–14*, *15–24*, *25–64*, and *65+*.
- 4. **Integrity Audit:** Integer conversions were verified with zero missing, NaN, or negative values across all 87,690 rows (3.45 MB total).

Attribute	Data Type	Description	Sample Value
<code>country</code>	String	Standard sovereign nation or territory name	India
<code>code</code>	String	ISO 3166-1 alpha-3 territory code	IND
<code>continent</code>	String	Standard continental classification (UN M49)	Asia
<code>year</code>	Integer	Observation year (1950 – 2023 continuous)	2023
<code>age_group</code>	String	Demographic cohort (<code>Under 5</code> , <code>5–14</code> , <code>15–24</code> , <code>25–64</code> , <code>65+</code>)	25–64
<code>population</code>	Integer	Official absolute population count	720,314,435

3. System Architecture & Performance Design

In accordance with assignment guidelines, the implementation uses **pure vanilla web technologies** without frameworks like React, Next.js, Angular, or backend database servers.

- **Client-Side Processing:** Ingests the 3.45 MB CSV using `d3.csv()` in ~35 milliseconds.
- **In-Memory Multi-Key Indexing:** Organizes rows upon load into JavaScript `Map` objects (`yearCountryMap` and `countryTimeSeries`). Filtering or scrubbing timeline years executes in under 2 milliseconds, maintaining 60 frames per second (FPS) responsiveness.
- **Responsive SVG Geometry:** SVG containers employ dynamic `viewBox` attributes combined with a debounced window `resize` listener to scale cleanly across desktop, laptop, and tablet displays.
- **Offline Redundancy:** Includes a local fallback script (`js/d3.v7.min.js`) that activates automatically if external CDN connections are unavailable during viva evaluations.

4. D3.js Visualization Design & Methodological Details

4.1 Interactive Population Bar Chart (Population by Country)

The bar chart illustrates country rankings for any selected year and continent. Horizontal bars are used because lengthy country labels (e.g., "Democratic Republic of the Congo") would become unreadable or require severe rotation in a vertical layout.

- **Scales:** `d3.scaleBand()` maps country names to vertical slots with 0.24 inner padding; `d3.scaleLinear()` maps population counts to horizontal pixel widths with `.nice()` tick boundaries.
- **Data Join Lifecycle:** Applies `.join(enter => ..., update => ..., exit => ...)` combined with `d3.transition().duration(500)` to animate bar dimensions, ranks, and numeric labels smoothly as filters change.
- **Dynamic Highlighting:** Selecting a country styles its bar with a distinctive warm amber color (`#d97706`) and bold typography.
- **Bidirectional Filtering:** Clicking any bar or Y-axis label selects that country across the entire dashboard.

4.2 Demographic Stacked Area Chart (Age Distribution Over Time)

The stacked area chart visualizes cohort evolution across the full 74-year timeline (1950 to 2023) for the selected country, defaulting to **India** as specified in assignment requirements.

- **Stack Computation:** `d3.stack().keys(['Under 5', '5-14', '15-24', '25-64', '65+'])` accumulates layer baselines across the five cohorts.
- **Cubic Interpolation:** `d3.area().curve(d3.curveMonotoneX)` generates smooth SVG paths without artificial overshoot.
- **Dual Modes (Absolute vs. Share %):**
 - *Absolute Mode:* Displays cumulative population growth in millions and billions.
 - *Share (%) Mode:* Normalizes annual cohort totals to 100%, directly illustrating the demographic transition from youth-heavy expansion to an aging structure.
- **Timeline Synchronization:** A vertical dashed reference line tracks the active slider year, while an interactive `d3.bisector` crosshair displays cohort breakdowns for any hovered year. Clicking the chart jumps the slider to that year.

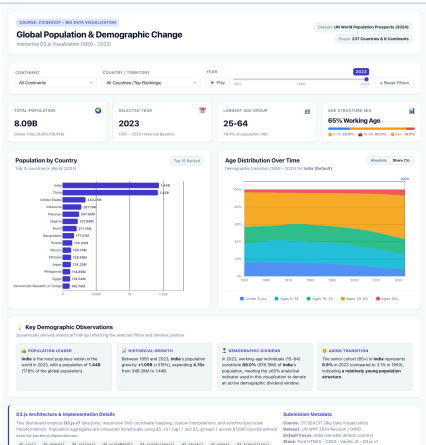


Figure 2: 100% Normalized Stacked Area Chart demonstrating India's demographic transition from 1950 to 2023. Notice the expansion of working-age cohorts (15–64, green & amber) and the gradual rise of older cohorts (65+, red).

5. Key Demographic Observations & Empirical Findings

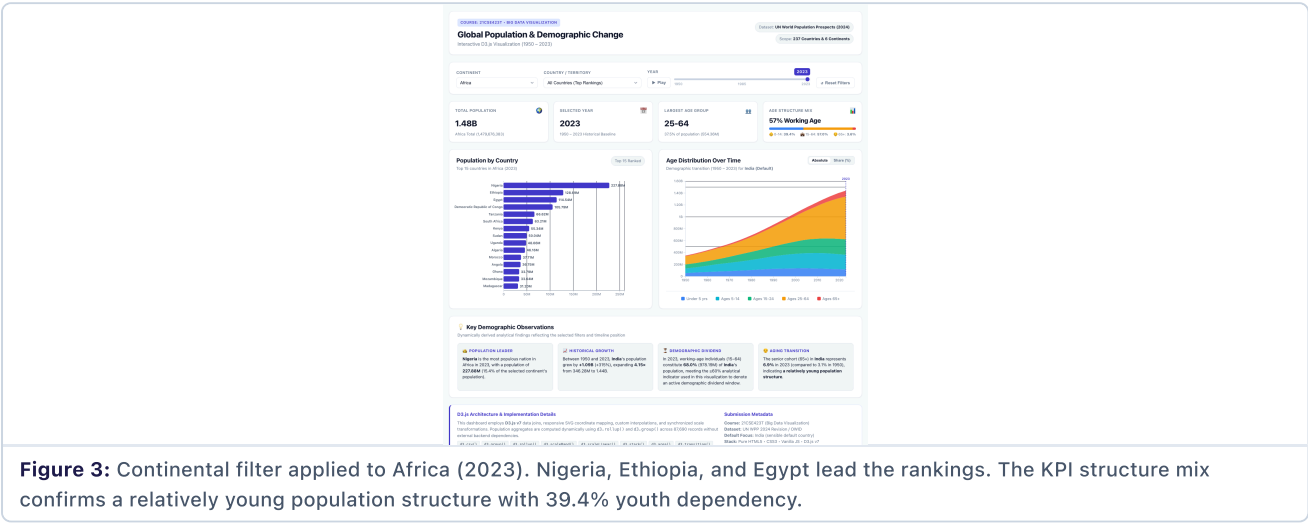
The dashboard features a dynamic analytical observations engine that derives conclusions in real time based on active filters.

5.1 Global Growth Trajectory (1950 vs. 2023)

World population expanded from **2.50 billion in 1950** to **8.09 billion in 2023** (a 3.24× expansion). In 2023, **India (1.438 billion)** officially overtook **China (1.423 billion)** to become the most populous nation, accounting for **17.8% of the global population**.

5.2 Continental Profile: Africa's Young Population Structure

Filtering the dashboard to **Africa** reveals a continental total of **1.48 billion** led by **Nigeria (227.88M)**, representing **15.4% of the selected continent's population**. Africa exhibits a distinctly young demographic structure: children (ages 0–14) comprise **39.4%** of the population, working-age individuals represent **57.0%**, and senior citizens (65+) account for just **3.6%**.

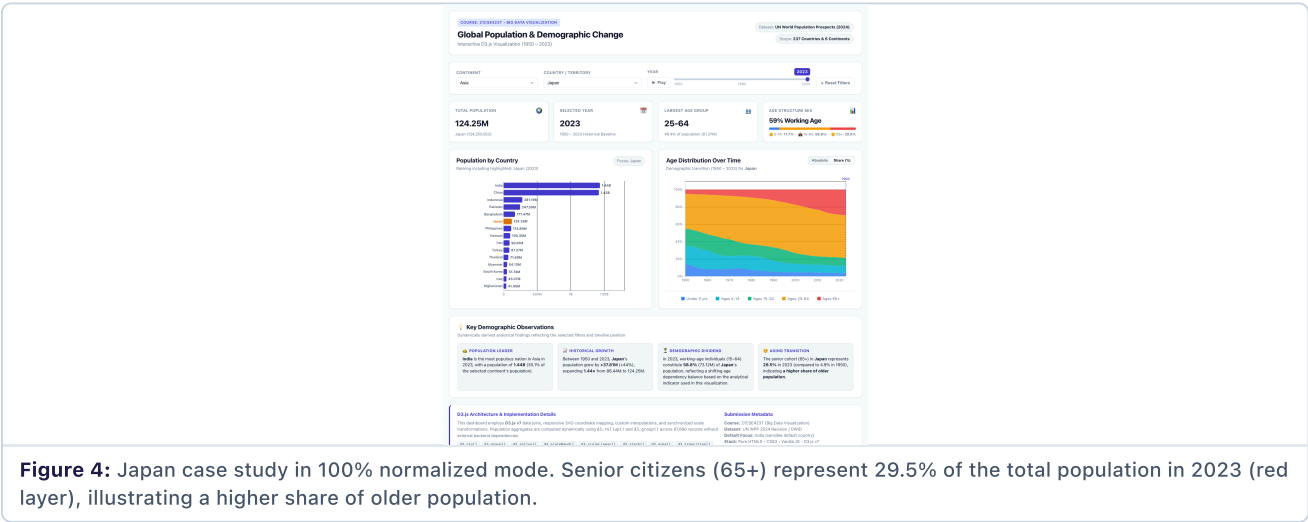


5.3 Demographic Dividend Indicator (India Case Study)

In 2023, working-age individuals (15–64) constitute **68.0% (978.19 million)** of India's population. This meets the $\geq 60\%$ analytical threshold indicator used in this visualization to denote an active *demographic dividend* window, where a high ratio of productive workers to dependents offers economic advantages.

5.4 Population Aging Trends (Japan Case Study)

In contrast, selecting **Japan** illustrates a pronounced population aging trend: the senior cohort (65+) increased from **4.9% in 1950** to **29.5% in 2023**, while children under 5 contracted to under **4%**, reflecting a higher share of older population and rising old-age dependency.



6. Longitudinal Analysis: 1970 Baseline Comparison

Dragging the timeline slider to **1970** allows historical comparison with the modern demographic landscape. In 1970:

- Global population was **3.69 billion** (less than half of the 2023 total).
- **China** was the clear global leader with **823.31 million (22.3% of the global population)**, followed by India (545.86M) and the United States (207.79M).
- European nations held prominent positions in the global top 15: Germany (78.2M), United Kingdom (55.7M), Italy (53.4M), and France (50.8M), nations that have since been superseded by faster-growing emerging economies.

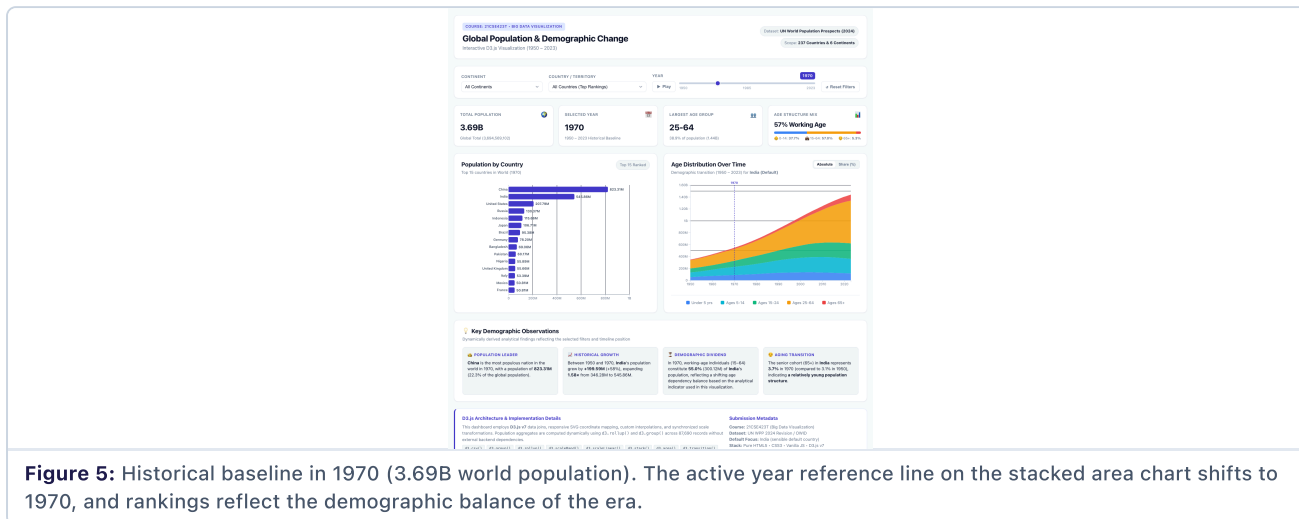


Figure 5: Historical baseline in 1970 (3.69B world population). The active year reference line on the stacked area chart shifts to 1970, and rankings reflect the demographic balance of the era.

7. Cloud Deployment & Live Hosting on Vercel

The complete project is deployed on **Vercel's global edge network** to enable seamless evaluation by mentors and examiners.

- **Live Production URL:** <https://big-data-visualization-one.vercel.app/>
- **Configuration (`vercel.json`):** Defines routing rules, enables CORS for open data sharing, and sets HTTP caching headers (`public`, `max-age=86400`, `stale-while-revalidate=604800`) for the 3.45 MB demographic dataset.
- **Continuous Deployment:** Connected directly to the GitHub repository (`TheCreativeCodeFlow/big-data-visualization`); any code updates automatically trigger optimized edge builds.
- **Deep-Linking Capability:** The live deployment supports URL parameters for direct evaluation of specific states (e.g., `?continent=Africa` or `?country=Japan&mode=pct`).

8. Viva Voce Defense & Oral Examination Guide

Q1: Why did you implement this in D3.js rather than using a high-level charting library?

Answer: High-level libraries (e.g., Chart.js, Highcharts) provide pre-built templates that conceal underlying coordinate mappings and data join lifecycles. D3.js operates directly on SVG elements and scales, providing granular control over mathematical coordinate systems, custom stacked baselines, transitions, and responsive refits. This demonstrates fundamental mastery of data visualization principles.

Q2: How does the stacked area chart compute coordinate baselines?

Answer: We invoke `d3.stack().keys(['Under 5', '5-14', '15-24', '25-64', '65+'])`. For every annual observation, D3 computes an interval $[y_0, y_1]$ where y_0 is the top of the previous layer and $y_1 = y_0 + \text{value}$. The area generator `d3.area().x(...).y0(...).y1(...)` transforms these numerical intervals into closed SVG polygon path strings using cubic monotone interpolation.

Q3: How did you ensure dynamic percentages sum strictly to 100.0%?

Answer: Independent rounding of decimal percentages frequently results in sums of 99.9% or 100.1%. We implemented the *Largest Remainder Method* (`roundPercentages`), which scales values to integer tenths, floors them, and distributes remaining tenths according to largest fractional remainders. This guarantees that all displayed cohort percentages sum strictly to 100.0%.

Q4: How do you achieve 60 FPS performance without a database backend?

Answer: The 87,690 records are loaded once via `d3.csv()` and indexed in-memory into JavaScript `Map` structures. When scrubbing the timeline slider or switching continents, the dashboard performs $O(1)$ dictionary lookups and filters an array of only 237 records, completing updates in under 2 milliseconds without network latency.

Q5: How do you distinguish between timeline slider state and tooltip hover state?

Answer: The area chart tooltip explicitly displays two distinct entries: **Hover Year:** (the year corresponding to current mouse cursor position) and **Timeline Slider:** (the global filter year), preventing user confusion when inspecting historical cohorts.

9. Conclusion

The **Global Population & Demographic Change Explorer** fulfills all requirements for the 21CSE423T Big Data Visualization assignment. Built with pure HTML5, CSS3, vanilla JavaScript, and D3.js v7, it presents an academically rigorous, aesthetically polished, and interactive exploration of modern demographic shifts. The application is maintained on GitHub and live on Vercel at <https://big-data-visualization-one.vercel.app/>.